

控制理論與飛行控制

第四週

Oct/23/2026

中原大學電機系

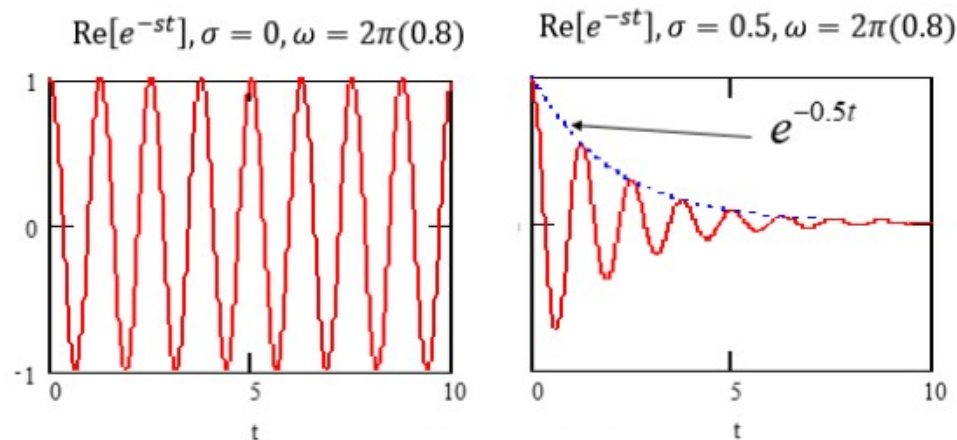
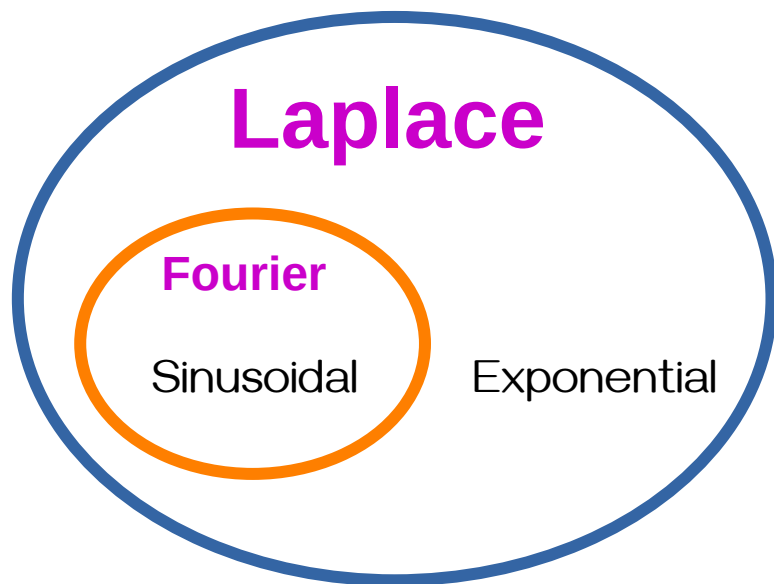
控制理論的數學工具

What is “s”?

s is a complex variable:

$$s = \sigma + j\omega$$

$$e^{-st} = e^{-(\sigma + j\omega)t} = e^{-\sigma t} e^{-j\omega t}$$



<https://www.youtube.com/watch?v=7UvtU75NXTg>

<https://www.youtube.com/watch?v=n2y7n6jw5d0>

Laplace Transform

$$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$$

$$\mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds = f(t)u(t)$$

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$e^{-at}u(t)$	$\frac{1}{s+a}$
$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

Differential Eq. Solved by LT

$$\frac{d^2y}{dt^2} + 12\frac{dy}{dt} + 32y = 32u(t)$$

$$s^2Y(s) + 12sY(s) + 32Y(s) = \frac{32}{s}$$

$$Y(s) = \frac{32}{s(s+4)(s+8)} = \frac{K_1}{s} + \frac{K_2}{(s+4)} + \frac{K_3}{(s+8)}$$

$$y(t) = (1 - 2e^{-4t} + e^{-8t})u(t)$$

The Transfer Function

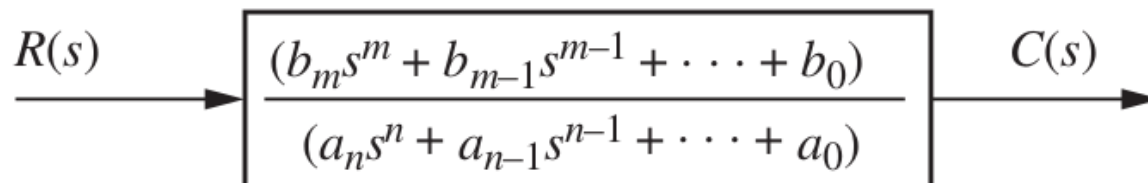
A general nth-order, linear, time-invariant differential equation :

$$a_n \frac{d^n c(t)}{dt^n} + a_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \dots + a_0 c(t) = b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \dots + b_0 r(t)$$

$c(t)$ is the output, $r(t)$ is the input

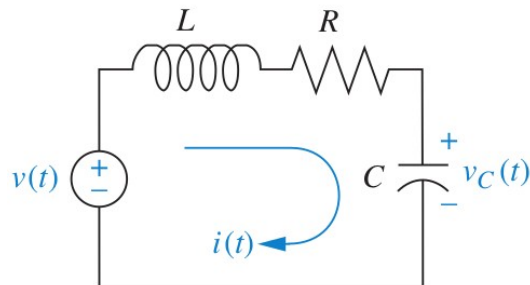
$$(a_n s^n + a_{n-1} s^{n-1} + \dots + a_0) C(s) = (b_m s^m + b_{m-1} s^{m-1} + \dots + b_0) R(s)$$

$$\frac{C(s)}{R(s)} = G(s) = \frac{(b_m s^m + b_{m-1} s^{m-1} + \dots + b_0)}{(a_n s^n + a_{n-1} s^{n-1} + \dots + a_0)}$$

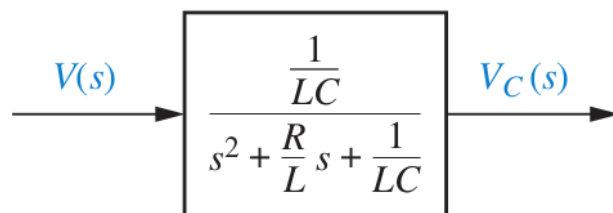


Electrical Network T.Func

Component	Voltage-current	Current-voltage	Voltage-charge	Impedance $Z(s) = V(s)/I(s)$	Admittance $Y(s) = I(s)/V(s)$
 Capacitor	$v(t) = \frac{1}{C} \int_0^1 i(\tau) d\tau$	$i(t) = C \frac{dv(t)}{dt}$	$v(t) = \frac{1}{C} q(t)$	$\frac{1}{Cs}$	Cs
 Resistor	$v(t) = Ri(t)$	$i(t) = \frac{1}{R} v(t)$	$v(t) = R \frac{dq(t)}{dt}$	R	$\frac{1}{R} = G$
 Inductor	$v(t) = L \frac{di(t)}{dt}$	$i(t) = \frac{1}{L} \int_0^1 v(\tau) d\tau$	$v(t) = L \frac{d^2 q(t)}{dt^2}$	Ls	$\frac{1}{Ls}$



$$L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int_0^t i(\tau) d\tau = v(t)$$



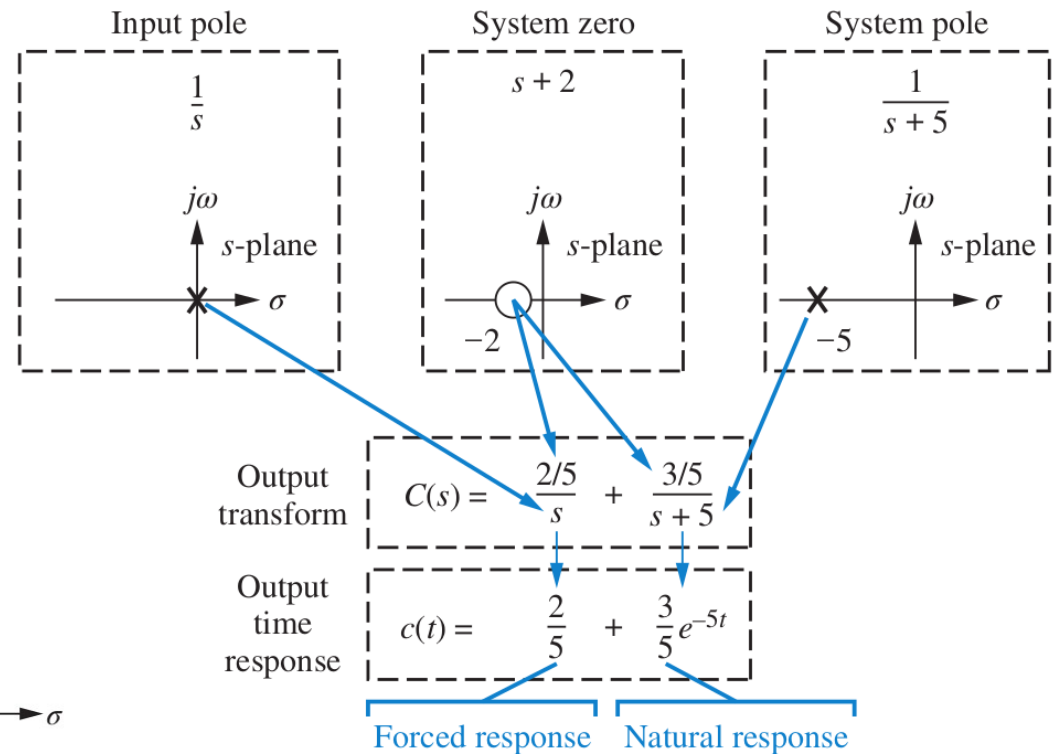
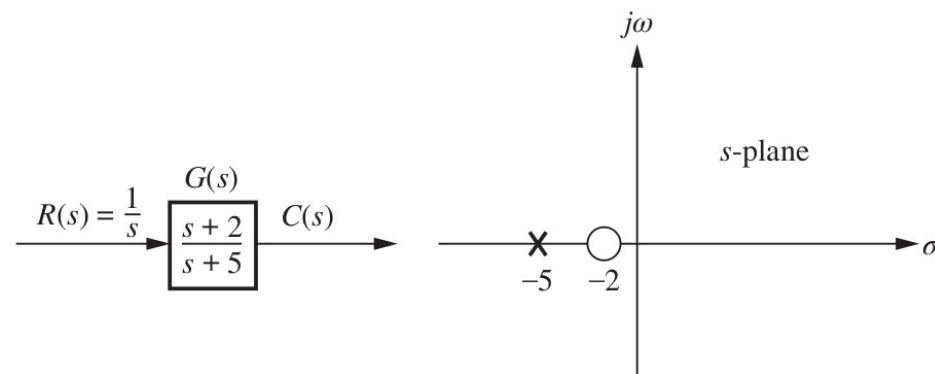
$$(LCs^2 + RCs + 1)V_C(s) = V(s)$$

Time Response

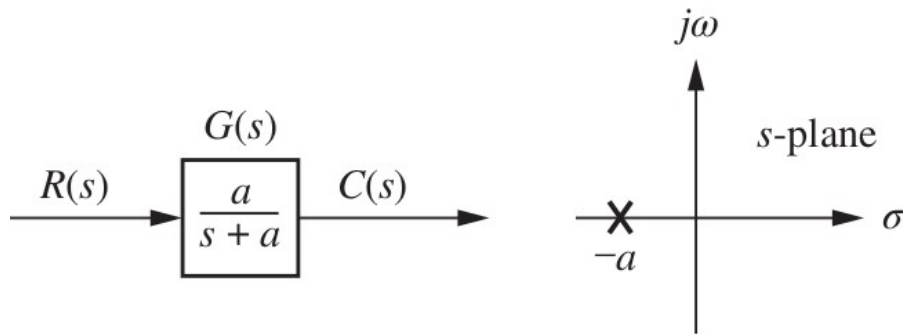
Poles and Zeros of a First-Order System

$$C(s) = \frac{(s+2)}{s(s+5)} = \frac{A}{s} + \frac{B}{s+5} = \frac{2/5}{s} + \frac{3/5}{s+5}$$

$$c(t) = \frac{2}{5} + \frac{3}{5}e^{-5t}$$



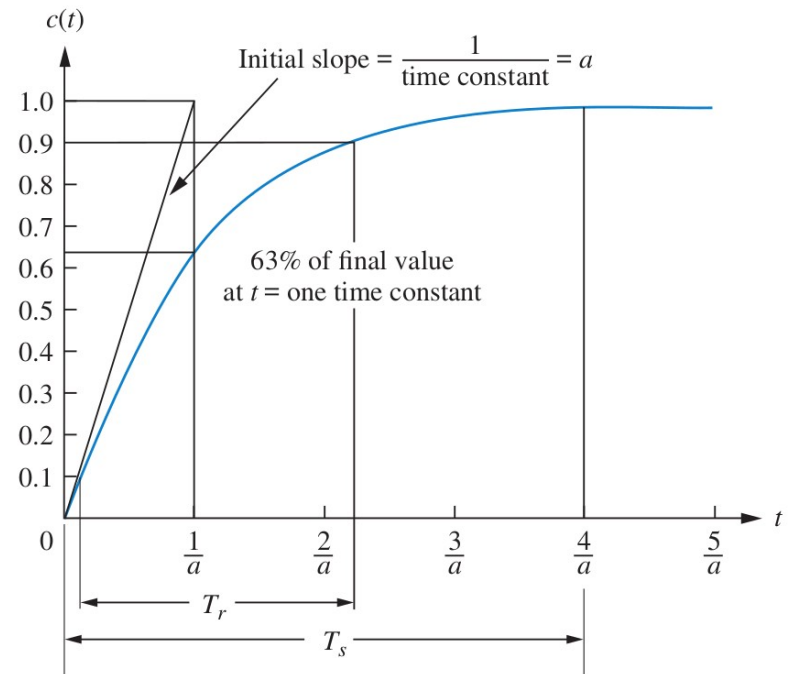
First-Order Systems



If the input is a unit step, where $R(s)=1/s$, the Laplace transform of the step response is $C(s)$

$$C(s) = R(s)G(s) = \frac{a}{s(s+a)}$$

$$c(t) = c_f(t) + c_n(t) = 1 - e^{-at}$$



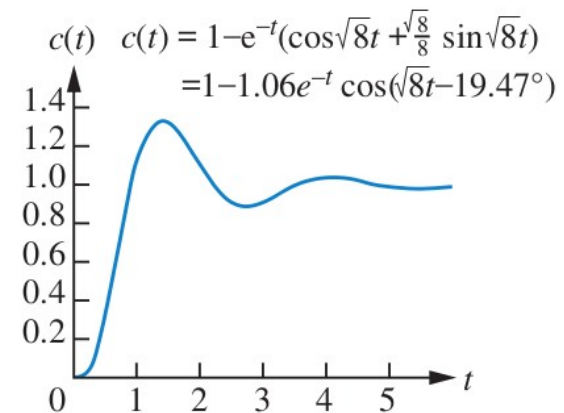
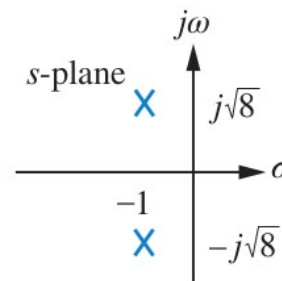
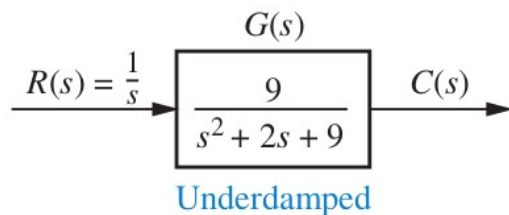
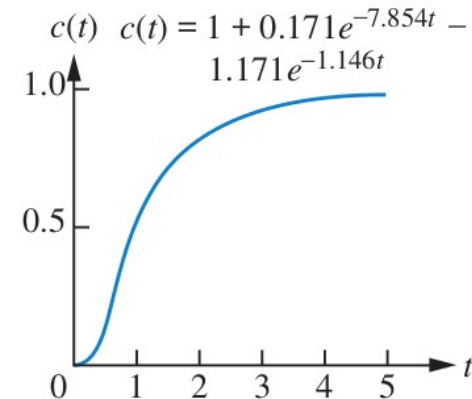
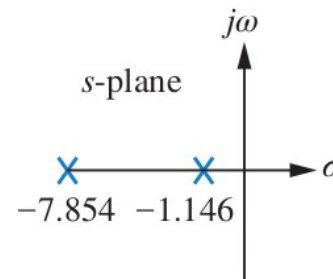
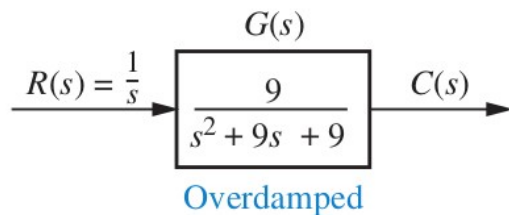
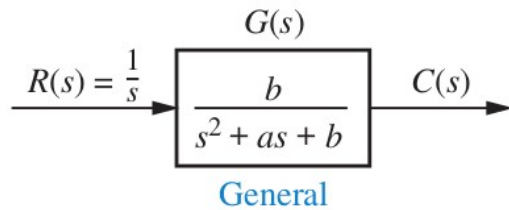
Rise Time, T_r

$$T_r = \frac{2.31}{a} - \frac{0.11}{a} = \frac{2.2}{a}$$

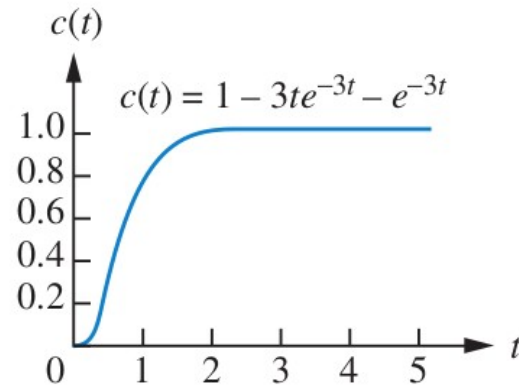
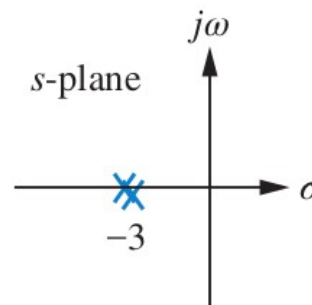
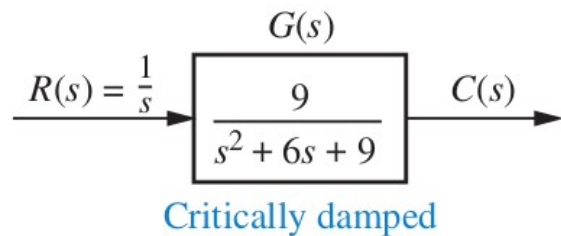
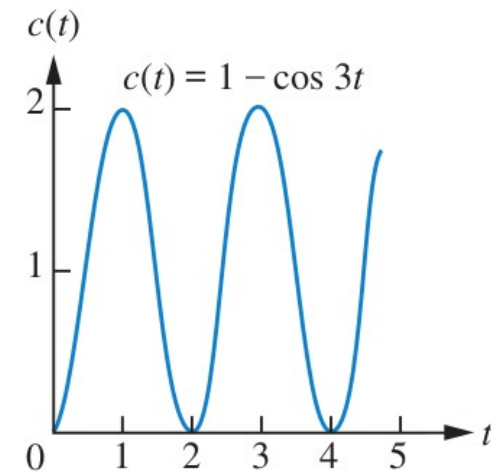
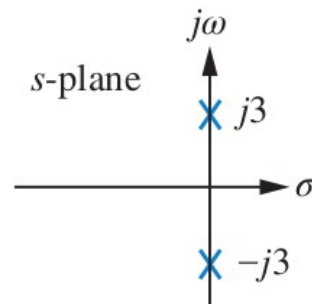
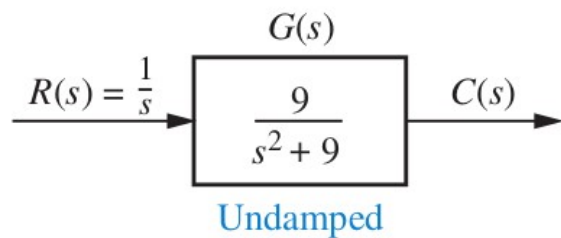
Settling Time, T_s

$$T_s = \frac{4}{a}$$

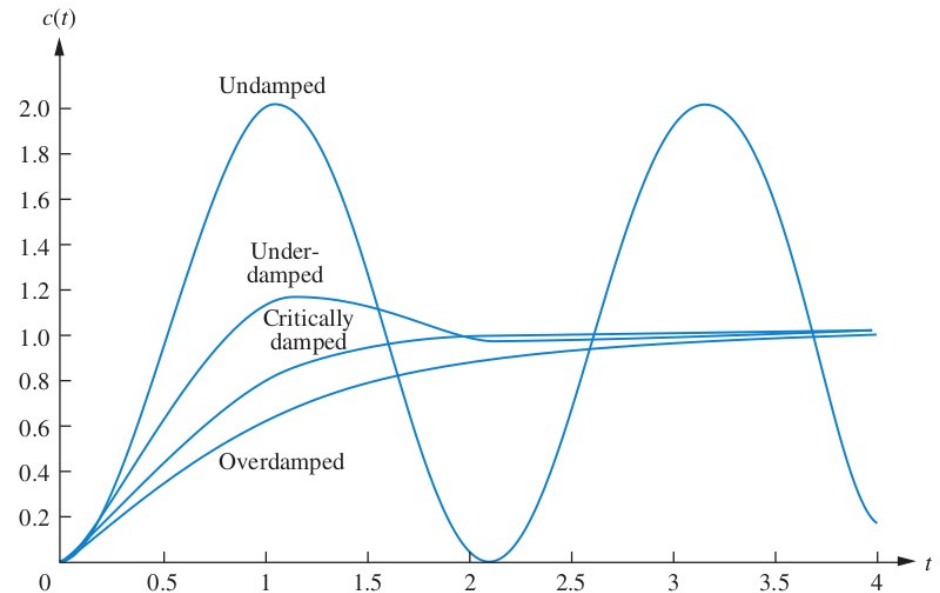
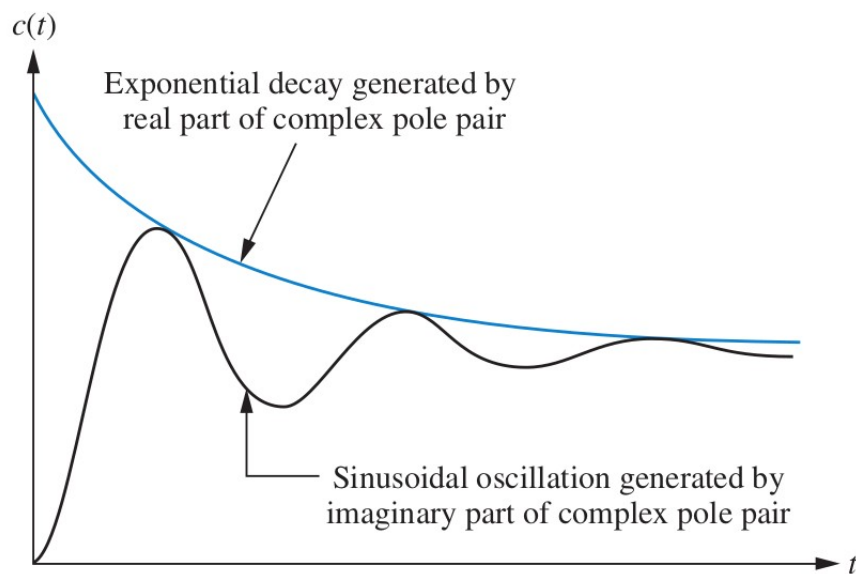
Second-Order Systems



Second-Order Systems



Second-Order Systems



暫態響應由一個指數衰減項（由系統極點的實部產生，決定其衰減振幅）和一個正弦波形（由系統極點的虛部產生）相乘組成。指數衰減的時間常數等於系統極點實部的倒數。

The General 2nd-Order System

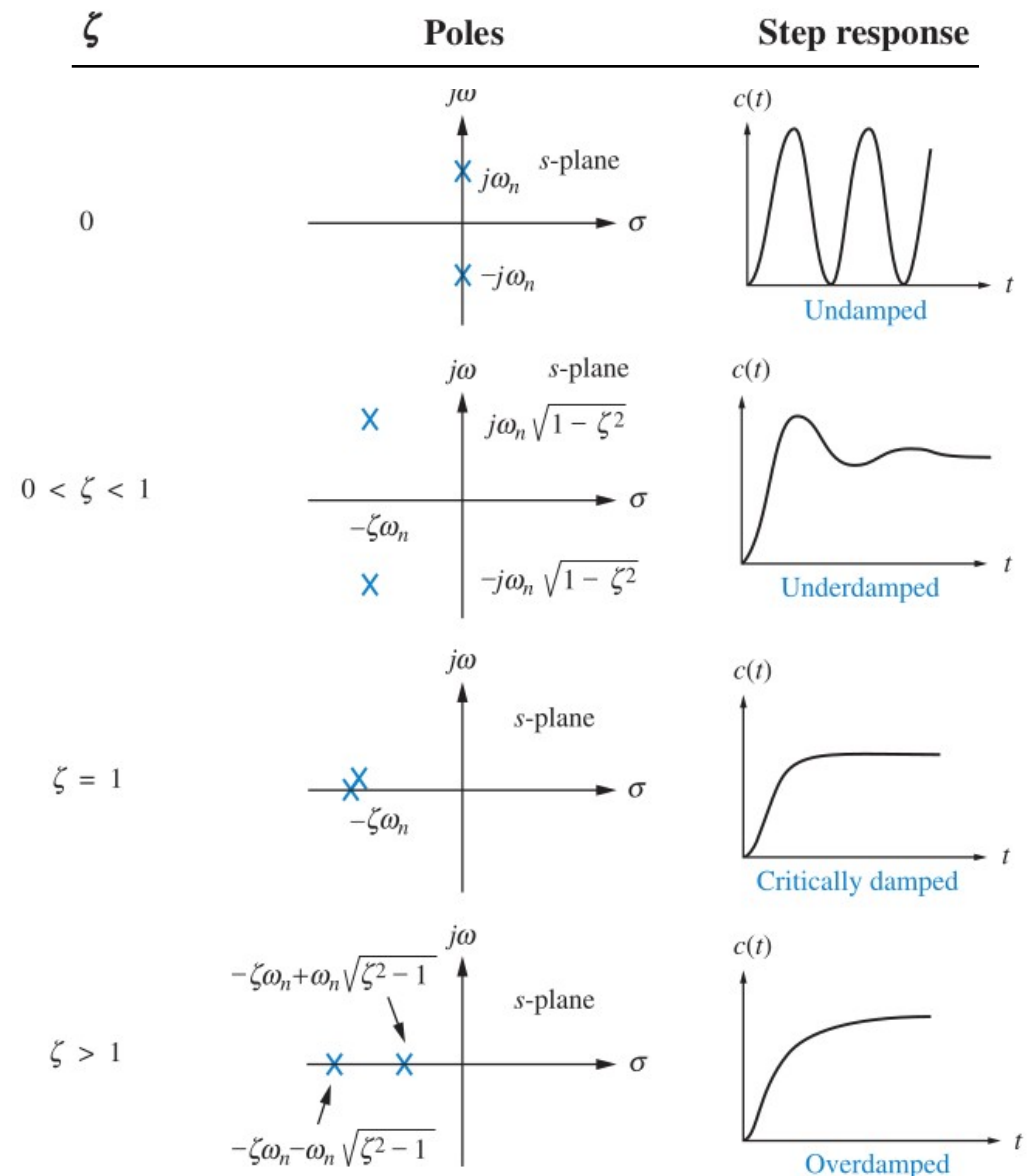
$$G(s) = \frac{b}{s^2 + as + b}$$

二階系統的自然頻率 ω_n 是指系統在沒有阻尼 (Damping) 情況下的震盪頻率

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

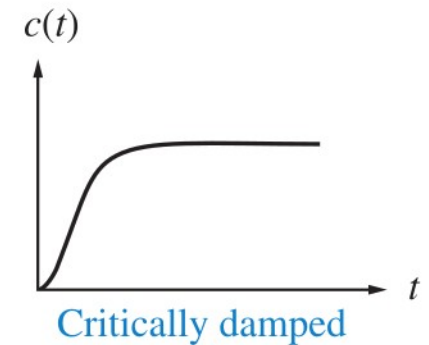
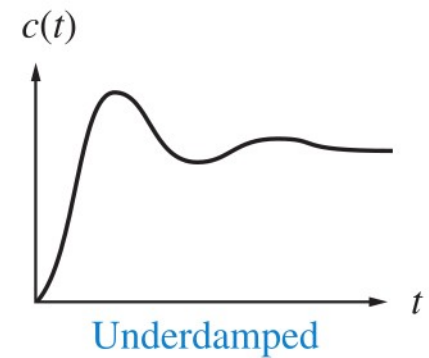
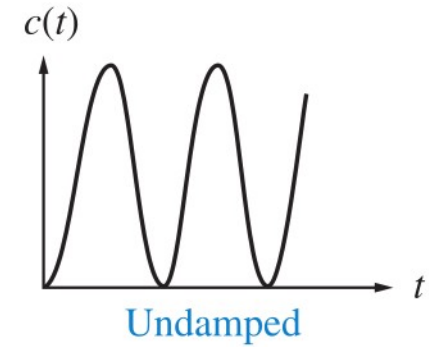
Damping Ratio

$$\zeta = \frac{\text{Exponential decay frequency}}{\text{Natural frequency (rad/second)}} = \frac{|\sigma|}{\omega_n} = \frac{a/2}{\omega_n}$$



2nd-Order System 練習題

$$G(s) = \frac{36}{s^2 + 4.2s + 36}$$



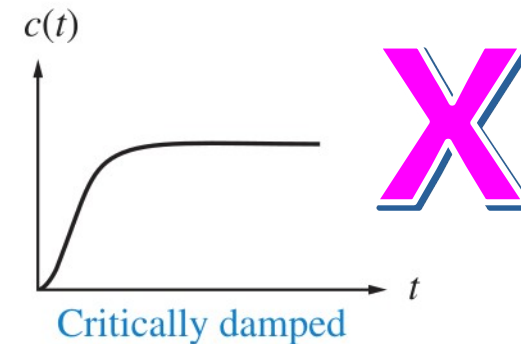
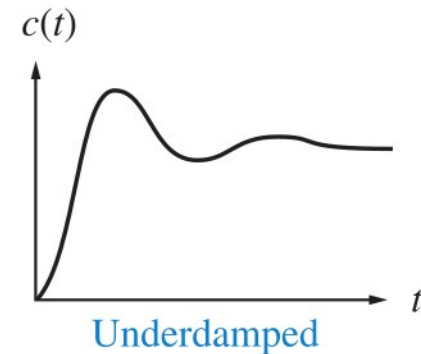
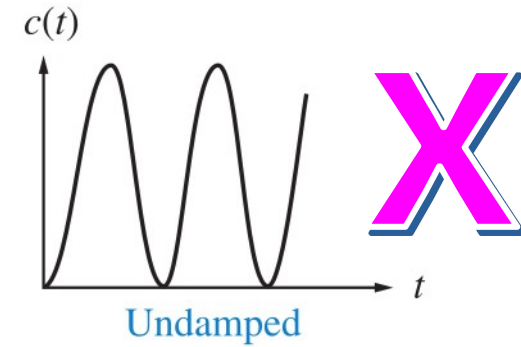
2nd-Order System 練習題

$$G(s) = \frac{36}{s^2 + 4.2s + 36}$$

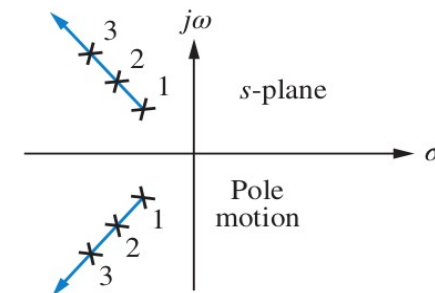
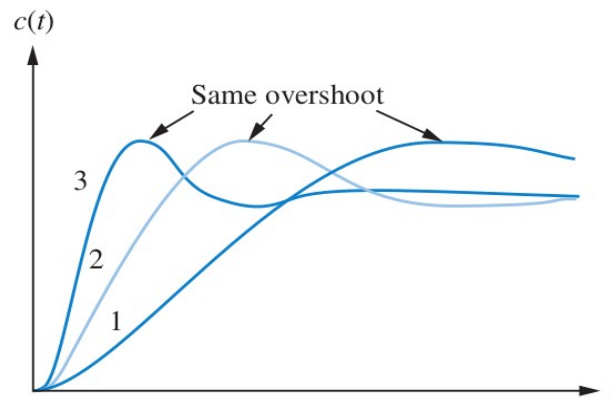
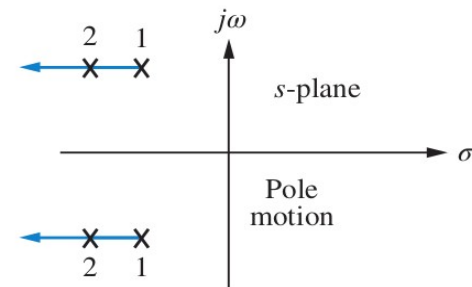
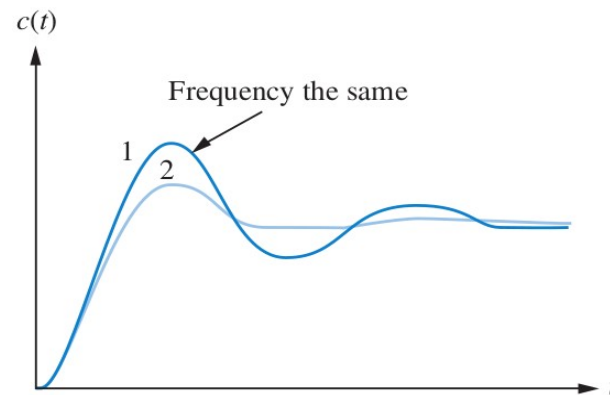
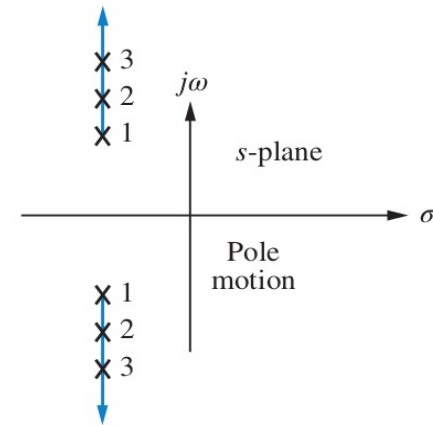
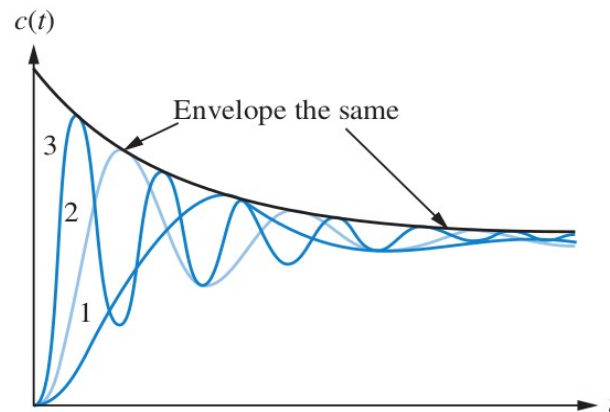
$$\omega_n^2 = 36 \Rightarrow \omega_n = 6$$

$$2\zeta\omega_n = 4.2 \Rightarrow \zeta = 0.35$$

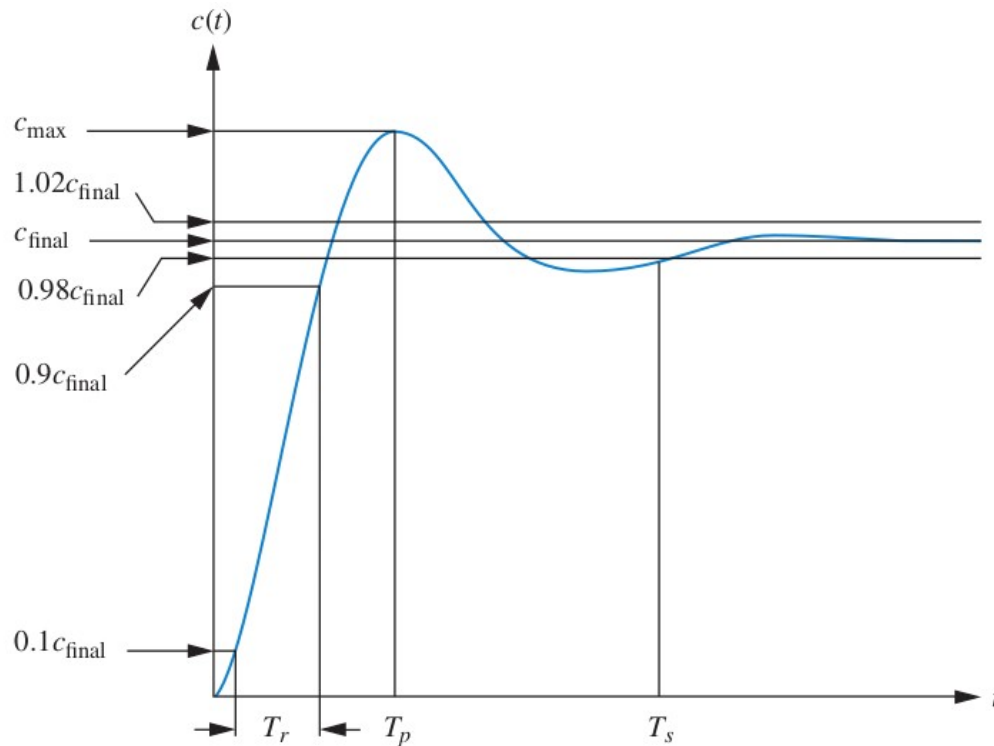
$$0 < \zeta < 1$$



The General 2nd-Order System



The General 2nd-Order System



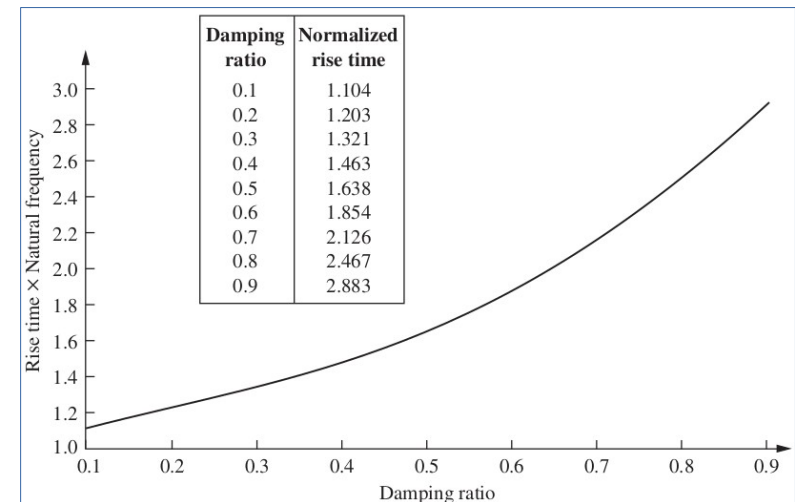
$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

$$T_s = \frac{4}{\zeta\omega_n}$$

$$c(t) = 1 - e^{-\zeta\omega_n t} \left(\cos \omega_n \sqrt{1 - \zeta^2} t + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin \omega_n \sqrt{1 - \zeta^2} t \right)$$

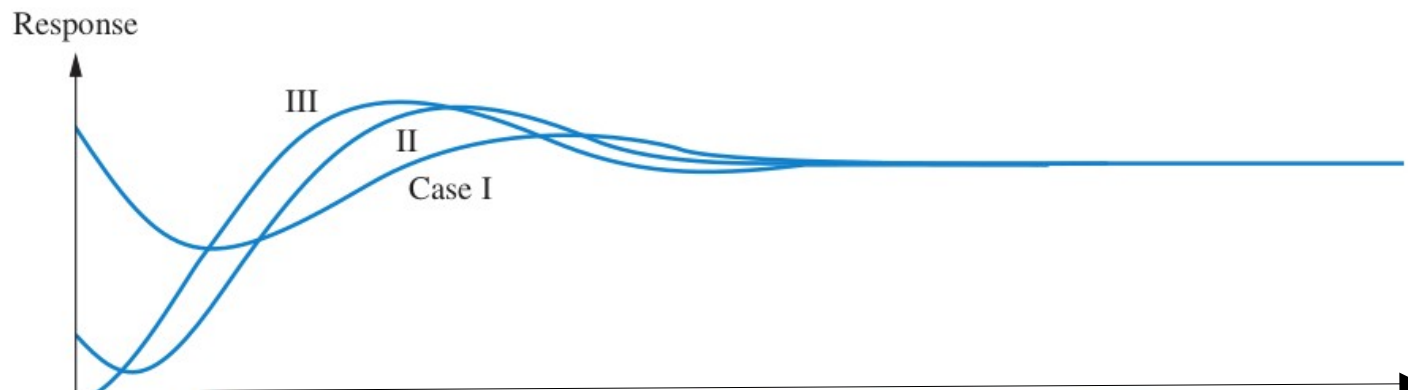
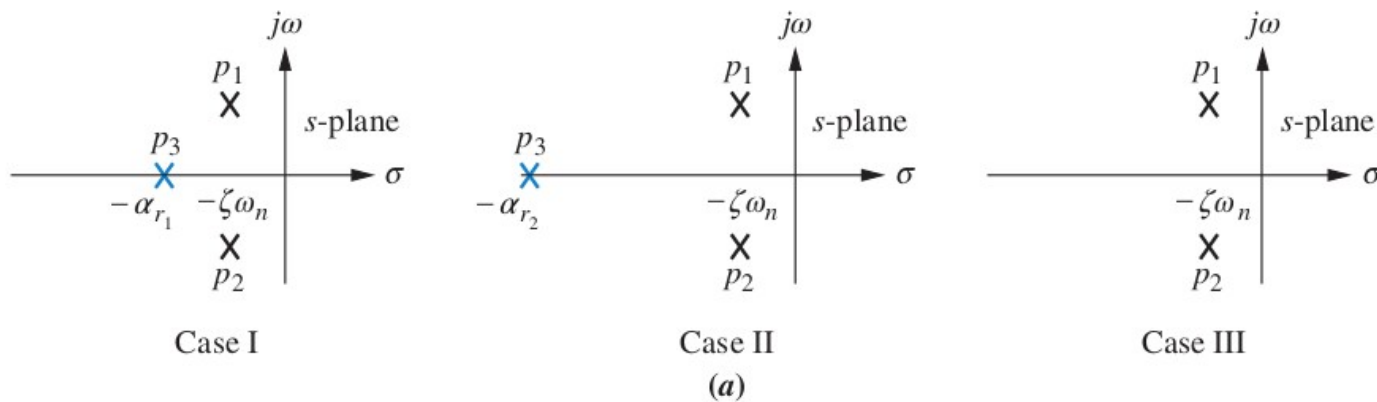
$$= 1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1 - \zeta^2} t - \phi)$$



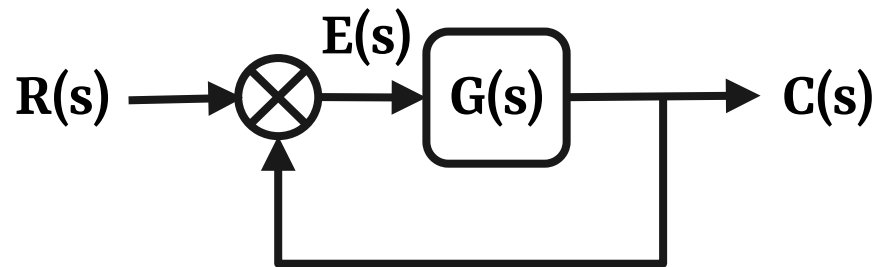
Response with Additional Poles

$$C(s) = \frac{A}{s} + \frac{B(s + \zeta\omega_n) + C\omega_d}{(s + \zeta\omega_n)^2 + \omega_d^2} + \frac{D}{s + \alpha_r}$$

$$c(t) = Au(t) + e^{-\zeta\omega_n t}(B \cos \omega_d t + C \sin \omega_d t) + De^{-\alpha_r t}$$



Stability



1. Error signal:

$$E(s) = R(s) - C(s)$$

2. Output signal:

$$C(s) = E(s) \cdot G(s)$$

3. Substitute $E(s)$ into $C(s)$:

$$C(s) = [R(s) - C(s)] G(s)$$

$$C(s) = R(s)G(s) - C(s)G(s)$$

4. Group $C(s)$ terms on the left:

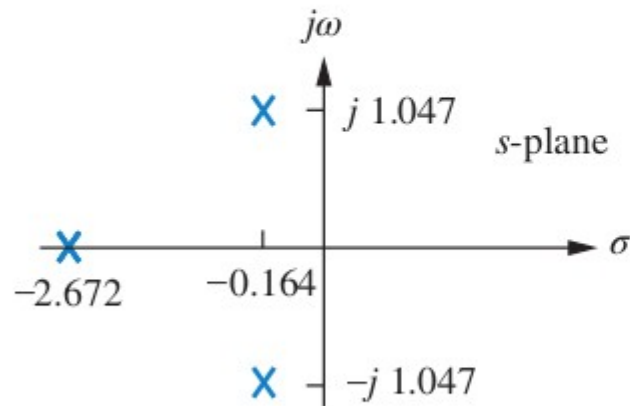
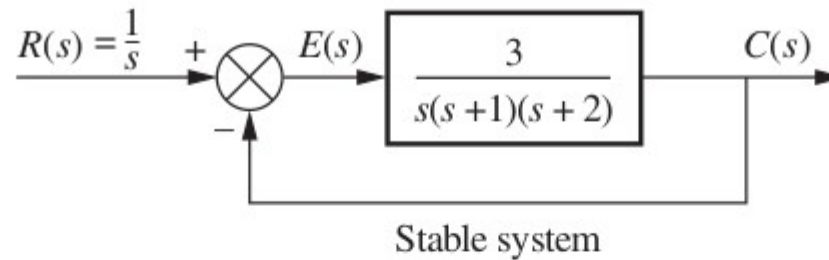
$$C(s) + C(s)G(s) = R(s)G(s)$$

$$C(s)[1 + G(s)] = R(s)G(s)$$

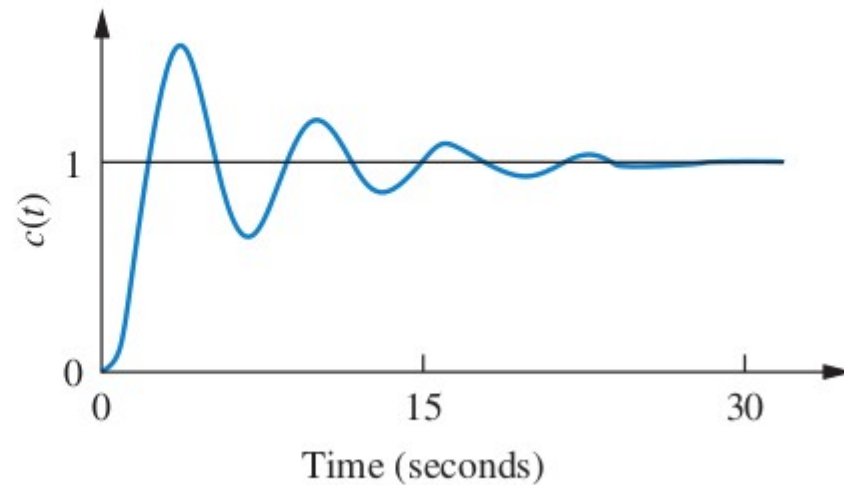
5. Solve for the ratio $\frac{C(s)}{R(s)}$:

$$T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

Stability

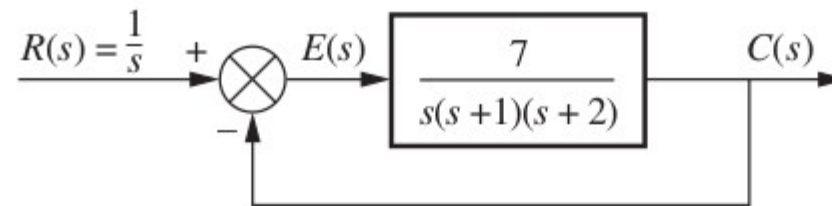


Stable system's
closed-loop poles
(not to scale)

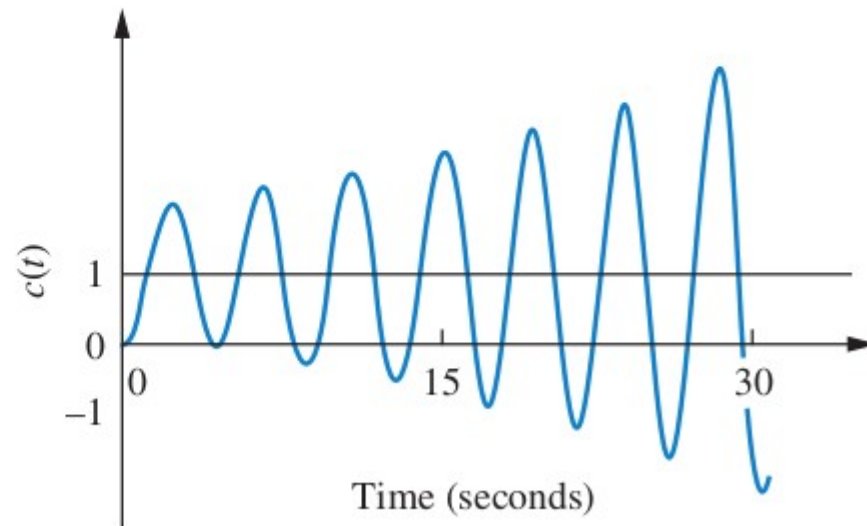
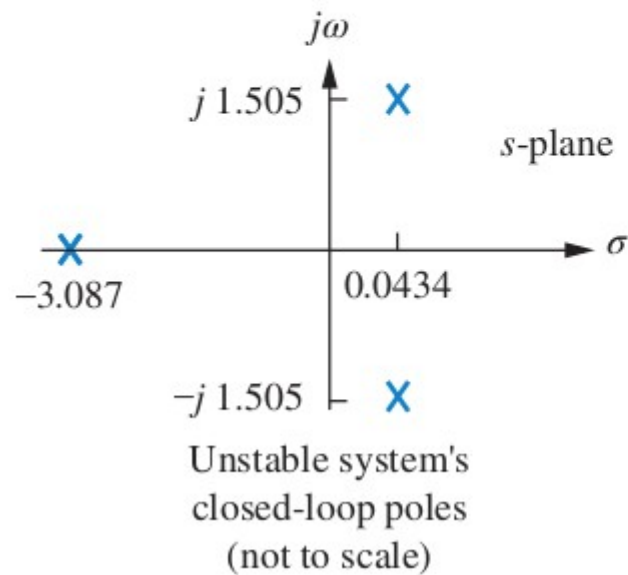


$$T(s) = \frac{3}{s(s+1)(s+2)+3}$$

Stability



Unstable system

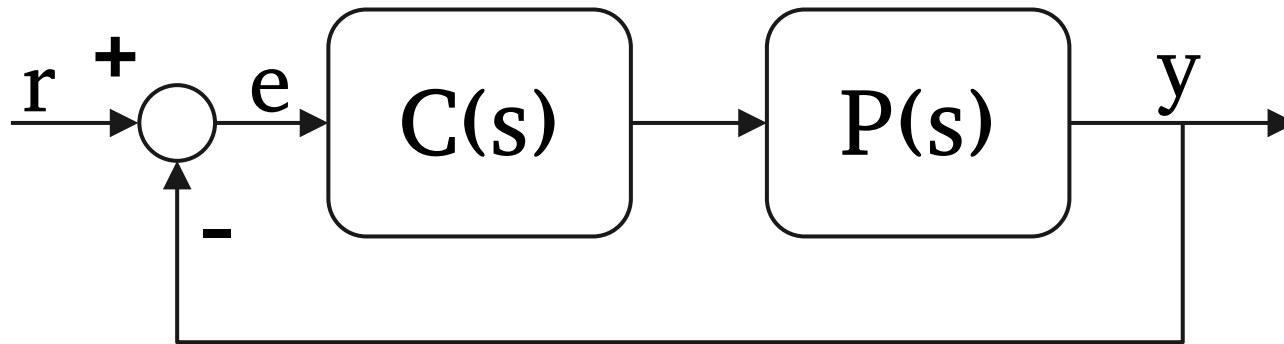


$$T(s) = \frac{7}{s(s+1)(s+2) + 7}$$

系統穩定性條件

極點位置	極點類型	系統穩定性
全在左半平面	任何類型	穩定
虛軸上	單根	邊緣穩定
虛軸上	重根	不穩定
右半平面	任何類型	不穩定

PID Controller



$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{d}{dt} e(t)$$

$$K_p + \frac{K_i}{s} + K_d s = \frac{K_p s + K_i + K_d s^2}{s}$$

PID Controller - K_p

$R(s)$: 參考輸入訊號 (Reference)

$Y(s)$ 或 $X(s)$: 系統輸出訊號 (Output displacement)

$E(s)$: 誤差訊號 (Error) , 定義為 $E(s) = R(s) - X(s)$

$C(s)$: 控制器傳遞函數 (Controller)

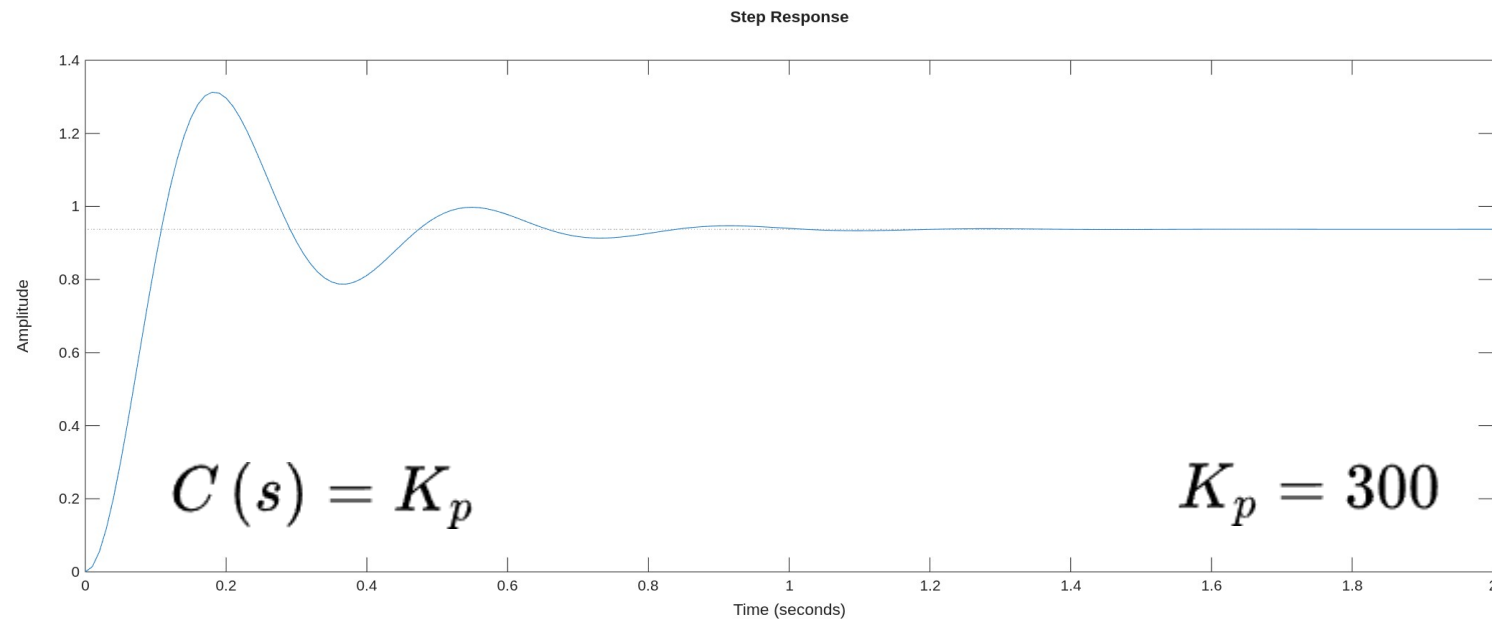
$P(s)$: 受控體傳遞函數 (Plant)

$$X(s) = E(s) \cdot C(s) \cdot P(s)$$

$$X(s) = [R(s) - X(s)] \cdot C(s) \cdot P(s)$$

$$T(s) = \frac{X(s)}{R(s)} = \frac{C(s)P(s)}{1 + C(s)P(s)}$$

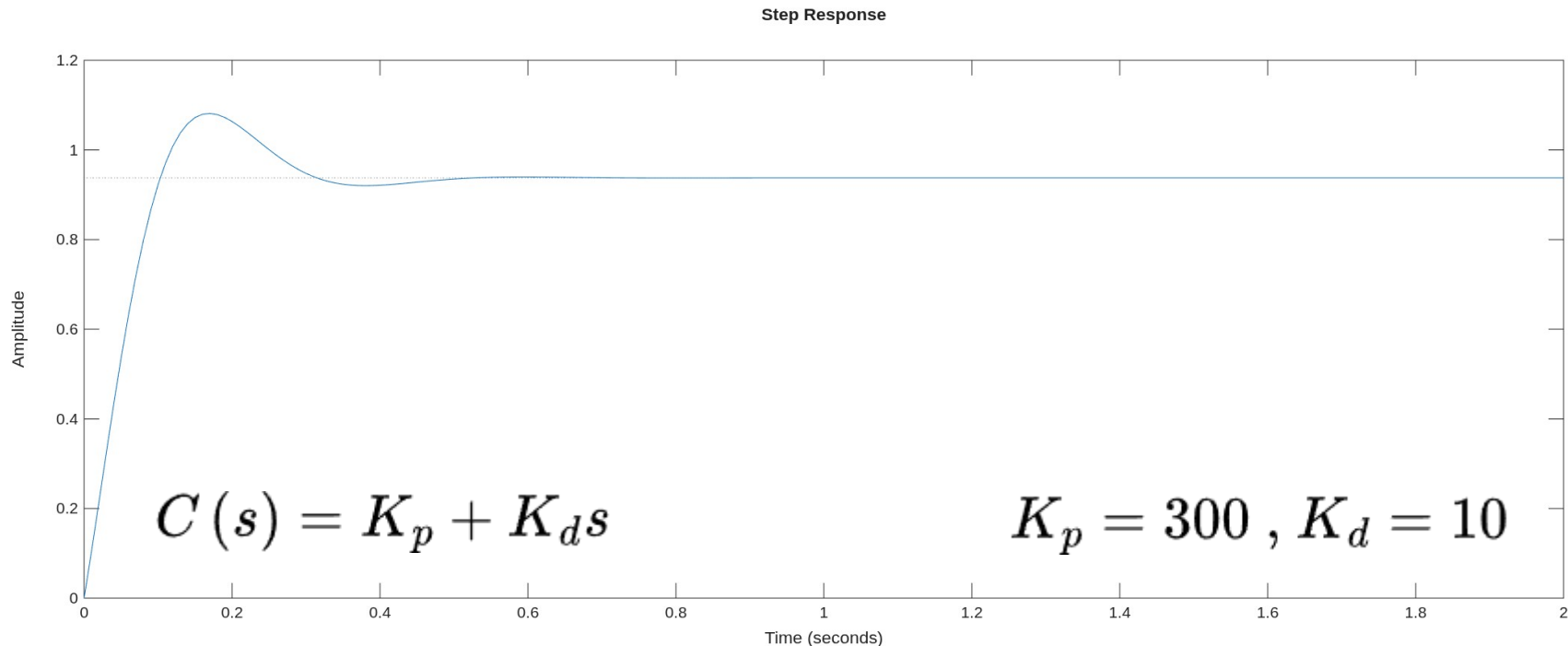
$$T(s) = \frac{X(s)}{R(s)} = \frac{K_p}{s^2 + 10s + (20 + K_p)}$$



PID Controller – K_p , K_d

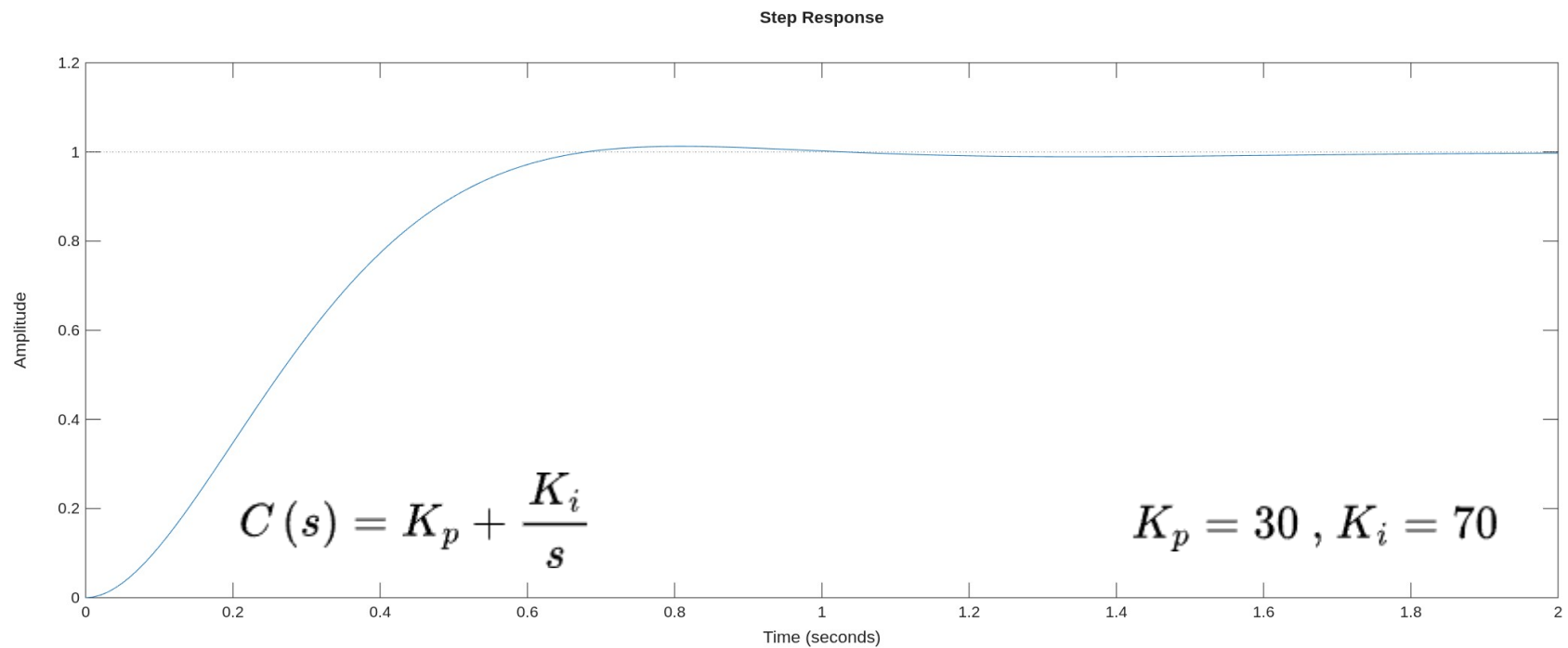
$$T(s) = \frac{K_d s + K_p}{s^2 + 10s + 20 + K_d s + K_p} = \frac{K_d s + K_p}{s^2 + (10 + K_d)s + (20 + K_p)}$$

$$K_p = 300, K_d = 10 \quad T(s) = \frac{10s + 300}{s^2 + 20s + 320}$$



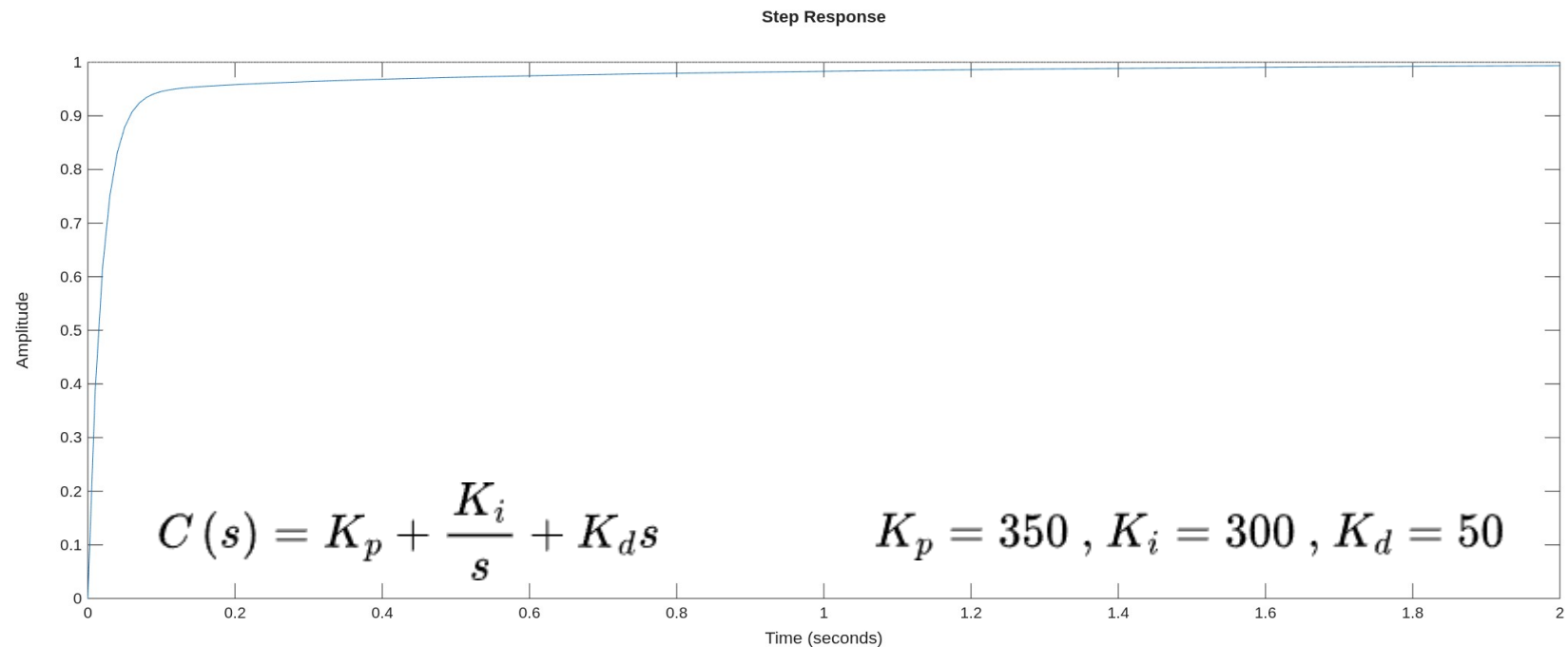
PID Controller – K_p , K_i

$$T(s) = \frac{X(s)}{R(s)} = \frac{K_p + \frac{K_i}{s}}{s^2 + 10s + 20 + K_p + \frac{K_i}{s}} = \frac{K_p s + K_i}{s^3 + 10s^2 + (20 + K_p)s + K_i}$$



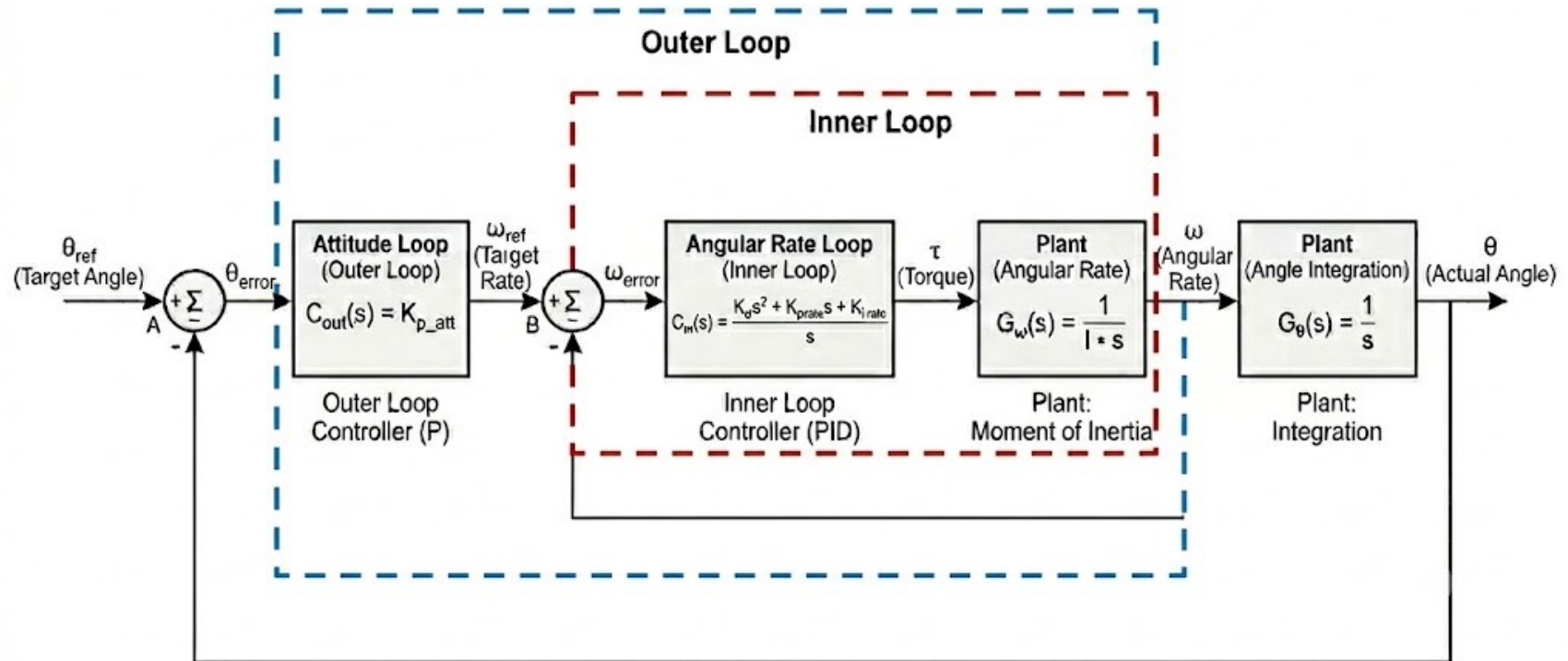
PID Controller – K_p , K_i , K_d

$$\begin{aligned} T(s) = \frac{X(s)}{R(s)} &= \frac{K_p + \frac{K_i}{s} + K_d s}{s^2 + 10s + 20 + K_p + \frac{K_i}{s} + K_d s} \\ &= \frac{K_d s^2 + K_p s + K_i}{s^3 + (10 + K_d)s^2 + (20 + K_p)s + K_i} \end{aligned}$$



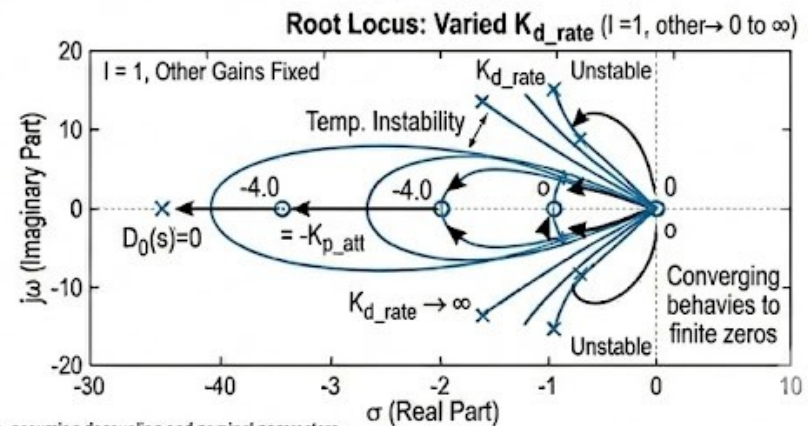
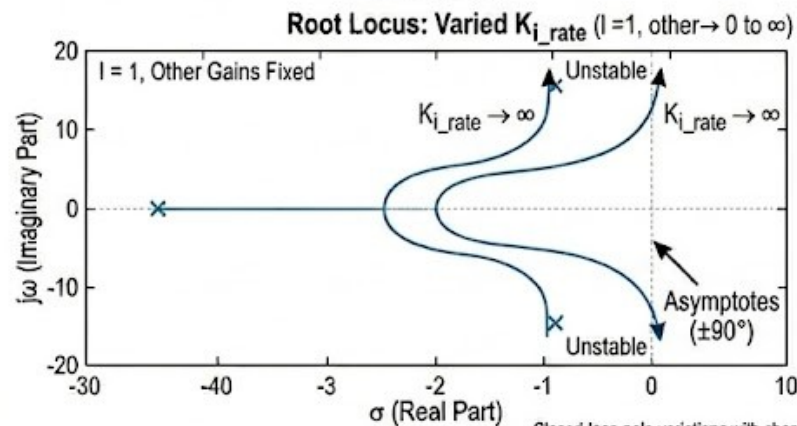
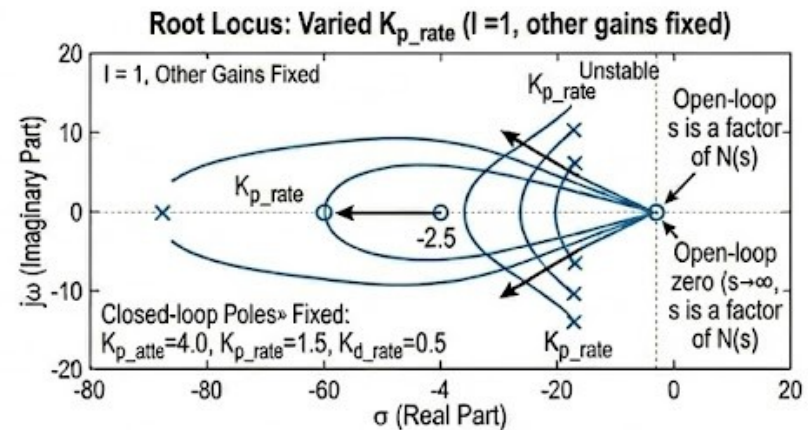
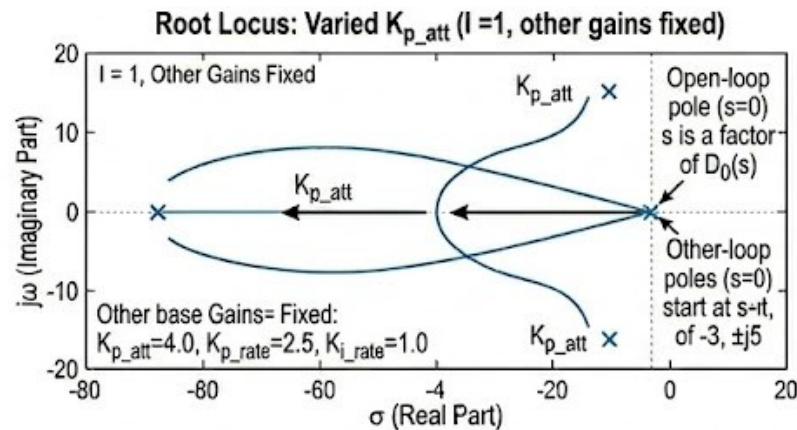
四軸無人機 - 雙環 PID 控制

QUADCOPTER PID DUAL-LOOP CONTROL SYSTEM BLOCK DIAGRAM (ATTITUDE & ANGULAR RATE)
Derived from Transfer Functions



雙環 PID 控制 – Root Locus

QUADCOPTER PID DUAL-LOOP SYSTEM: s-PLANE ROOT LOCUS ANALYSIS



Closed-loop pole variations with changing gains, assuming decoupling and nominal parameters.